
Data Visualization GUI Interactivity Study

Booking (Ahead of time)

- Send consent form

Checklist (Preparation Before Study)

- A. 3 Copies Consent Form
- B. Charge any needed devices
- C. 2 Pens
- D. Verify with each of the X participants:
 - Time and place of meeting
 - Contact info:
 - Phone number
 - Address
 - Who will be there
 - Go over any consent questions if lingering
- E. Surveys:
 - 2 background questionnaires

Setup

- Set up design materials
 - Assign the participant an anonymous ID
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Consent (~10min)

Introduction:

Welcome! Thank you for participating in this study. Throughout the study, I may need to read from a script to ensure that I keep our session on track.

The purpose of this study is to investigate user interactions with dashboard widgets, graphs, the carousel, sidebars, and the overall data visualization pipeline in a web-based AI-driven platform. By studying how participants engage with these features, we aim to improve their design and usability.

This study is being conducted by myself, George Morales.

Important Information

Many of the study tasks involve answering questions about your background and experience with data visualization tools. You will also be asked to interact with various dashboard elements, such as uploading datasets, labeling columns, generating graphs, customizing widgets, and navigating the carousel. Additionally, we will conduct a semi-structured interview to discuss your experience, challenges, and preferences.

Your participation is entirely voluntary, and you may choose to withdraw at any time. If you have any questions before we begin, feel free to ask.

Study Procedure

Today's session consists of two main activities.

First, I will ask you to upload a dataset, describe it, and label its columns for visualization.

Next, I will introduce you to the dashboard workspace, where you can interact with data visualizations, graphs, and charts. You will have the opportunity to explore the interface, adjust settings, and experiment with different configurations to see how the system responds.

Then, I will ask you to create and customize dashboard widgets, rearrange elements within the carousel, and interact with the sidebar to refine the display of your visualizations.

Finally, I will ask you to provide feedback on your experience, share any challenges or preferences you encountered, and discuss potential improvements to the interaction design.

Design Presentation (~5-10min.)

This will be done on my laptop. The browser-based GUI design consists of two main parts: data upload and annotation and dashboard interactions. Users start by uploading a dataset (CSV/Excel), labeling columns, and adding descriptions for context. Once processed, they enter the interactive dashboard, where they can generate and customize graphs, explore widgets, and navigate visualizations within a carousel. Sidebars provide additional options for filtering and refining insights.

Interview (~15min.)

I'd like to ask you a few questions about your experience interacting with the dashboard, including the data upload process, annotation, and visualization features. Please share your thoughts as openly as possible.

1. ***How intuitive was the data upload and annotation process? Were there any challenges or areas for improvement?***

The data upload and annotation process felt mostly intuitive, especially the way column annotation was handled—it was clear and easy to follow. However, having to manually select a data type for each column became a bit tedious, especially for datasets with many different columns. It would be helpful if the system could suggest or auto-detect data types to reduce the manual effort required.

2. ***How easy was it to navigate and interact with the dashboard, widgets, graphs, and carousel?***

Navigating and interacting with the dashboard felt straightforward, and the overall layout was easy to understand. However, the widgets were static, meaning they couldn't be resized or repositioned, which limited customization. Additionally, the carousel couldn't be hidden, which sometimes felt restrictive when focusing on a specific visualization.

3. ***Were the customization options sufficient for refining visualizations? If not, what additional features would you find useful?***

The customization options were solid, especially with the widget info section, the ability to expand widgets to full screen. Also, the hover interactions for the data points on the graphs were neat. These features made it easy to explore and understand the data behind the visualizations. However, more advanced customization options, like axis scaling, label placements, or applying different color schemes, could improve the overall interaction GUI.

4. ***Did anything about the interface feel confusing, frustrating, or particularly helpful? Why?***

The interface was intuitive, with a smooth flow from data upload to visualization. Expanding widgets and hovering over data points made analysis easier. However, manually selecting data types for each column could be tedious for larger datasets.

[My Notes]

- *Interface flow is intuitive and smooth.*
- *Widget expansion and hover interactions are useful.*
- *Manually selecting data types for each column is tedious for large datasets.*
- *Auto-detection or bulk selection for data types would improve efficiency.*
- *Fixed carousel can make the interface feel crowded; adding a hide option could help.*

Conclusion

Thank you for your help with our study. We are finished with all the session's tasks.